

Project Documentation

Laptop Price Prediction using Machine Learning

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Laptop Price Prediction using Machine Learning

1. Introduction

Machine Learning is one of the most important fields of Artificial Intelligence that allows computers to learn from data and make predictions. In this project, Machine Learning algorithms are used to predict laptop prices based on different laptop specifications such as RAM, Processor, Storage, GPU, Brand, and Screen Features.

The base research paper used for this project was:

“Forecasting Laptop Prices: A Comparative Study of Machine Learning Algorithms for Predictive Modeling” by Venkata Ravi Kiran Kolla.

The research paper focused mainly on laptop price prediction using Machine Learning models. We used this paper as a base idea and improved the project by adding multiple Machine Learning algorithms, comparison analysis, preprocessing techniques, visualization, and performance evaluation.

This project helps users and companies estimate laptop prices more accurately using Machine Learning techniques.

2. Problem Statement

Laptop prices depend on many factors such as:

- Brand
- RAM
- Processor
- GPU
- Storage Type
- Screen Size

- SSD/HDD
- Operating System

Manually predicting laptop prices is difficult because many specifications affect the final price. Traditional methods may not provide accurate results.

The main problem addressed in this project is:

“How to accurately predict laptop prices using Machine Learning algorithms?”

3. Scope

The scope of this project includes:

- Predicting laptop prices using Machine Learning
- Comparing different Machine Learning models
- Finding the best-performing algorithm
- Improving prediction accuracy
- Using preprocessing techniques for better results
- Visualizing data through graphs and charts
- Applying advanced algorithms such as XGBoost and SVM

This project can be useful for:

- Online laptop stores
- E-commerce websites
- Customers comparing laptop prices
- Companies analyzing market trends

4. Objectives

The main objectives of this project are:

1. To develop a Machine Learning-based laptop price prediction system.
2. To preprocess and clean the dataset for accurate predictions.
3. To apply multiple Machine Learning algorithms.
4. To compare model performances using accuracy scores.
5. To identify the best-performing model.
6. To visualize the dataset using graphs and charts.
7. To improve the base research paper by adding extra features and detailed analysis.

5. Solution of Problem

To solve the laptop price prediction problem, the following steps were performed:

Step 1: Data Collection

The laptop dataset was collected containing features such as:

- Company Name
- Laptop Type
- RAM
- CPU
- GPU
- Storage
- Weight
- Operating System
- Price

Step 2: Data Preprocessing

The dataset was cleaned and processed using:

- Removing unnecessary columns
- Handling missing values
- Label Encoding

- Feature extraction
- Data scaling

Step 3: Model Training

Different Machine Learning models were applied:

- Logistic Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- AdaBoost
- XGBoost
- K-Nearest Neighbors (KNN)
- Support Vector Machine (SVM)

Step 4: Model Evaluation

Models were evaluated using:

- Accuracy Score
- Confusion Matrix
- Classification Report
- Performance Comparison

Step 5: Best Model Selection

After comparison, the model with the highest accuracy was selected.

The best-performing model was:

Support Vector Machine (SVM) with approximately: 95% Accuracy

6. Tabular Form Result

Model	Accuracy
Logistic Regression	Lower Accuracy
Decision Tree	Good Performance
Random Forest	84.26%
Gradient Boosting	84.78%
XGBoost	85.37%
KNN	Moderate Performance
SVM	95.35% (Highest)

7. Visualization

Different visualizations were used to better understand the dataset and results.

Bar Chart

Bar charts were used to:

- Compare model accuracies
- Compare laptop brands
- Analyze feature importance

Scatter Plot

Scatter plots were used to:

- Observe the relationship between laptop specifications and price
- Analyze patterns in the dataset

Heatmap

Heatmaps helped in:

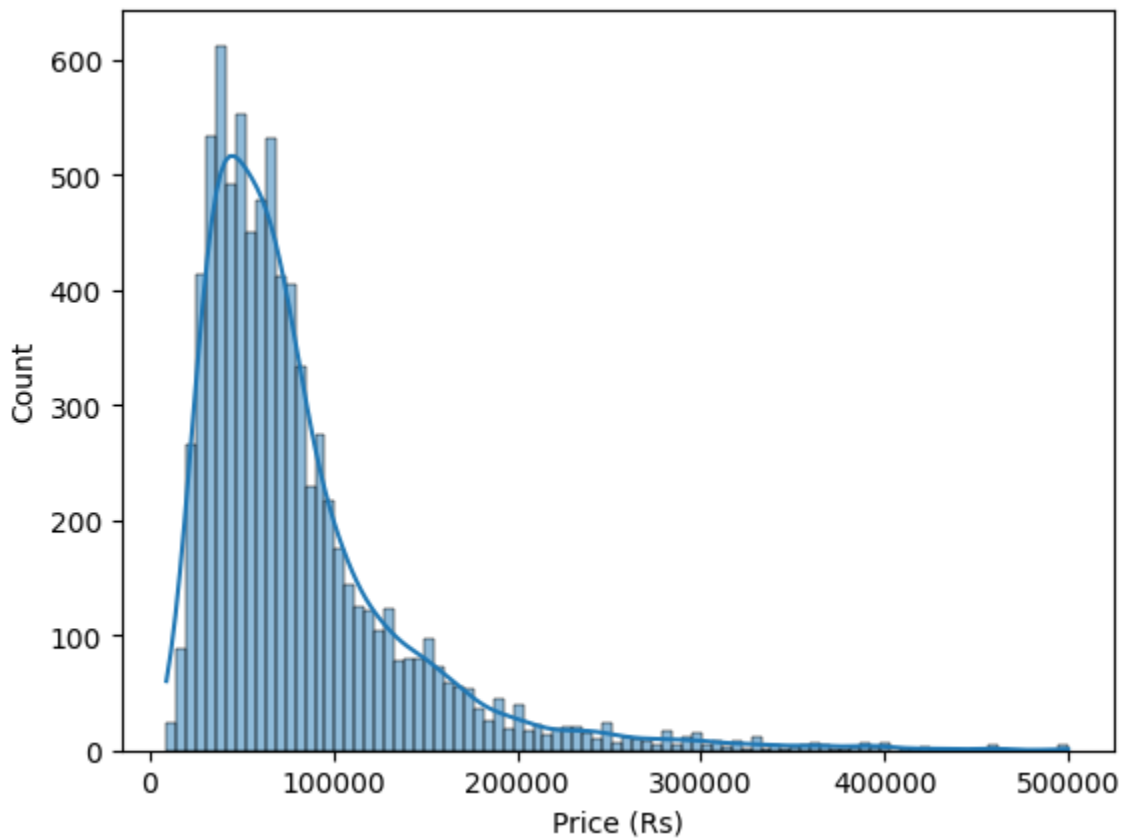
- Understanding feature correlation
- Identifying important features affecting laptop prices

Comparative Graphs

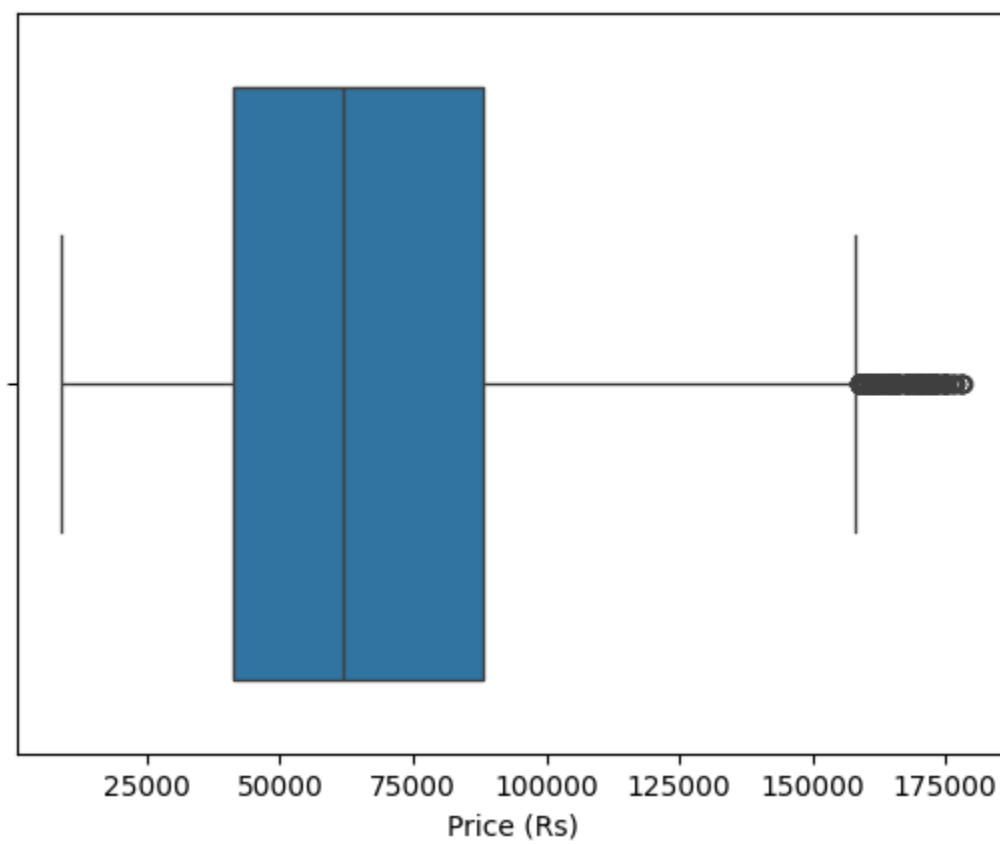
Graphs were used to compare:

- Accuracy of Machine Learning models
- Performance of algorithms
- Prediction efficiency

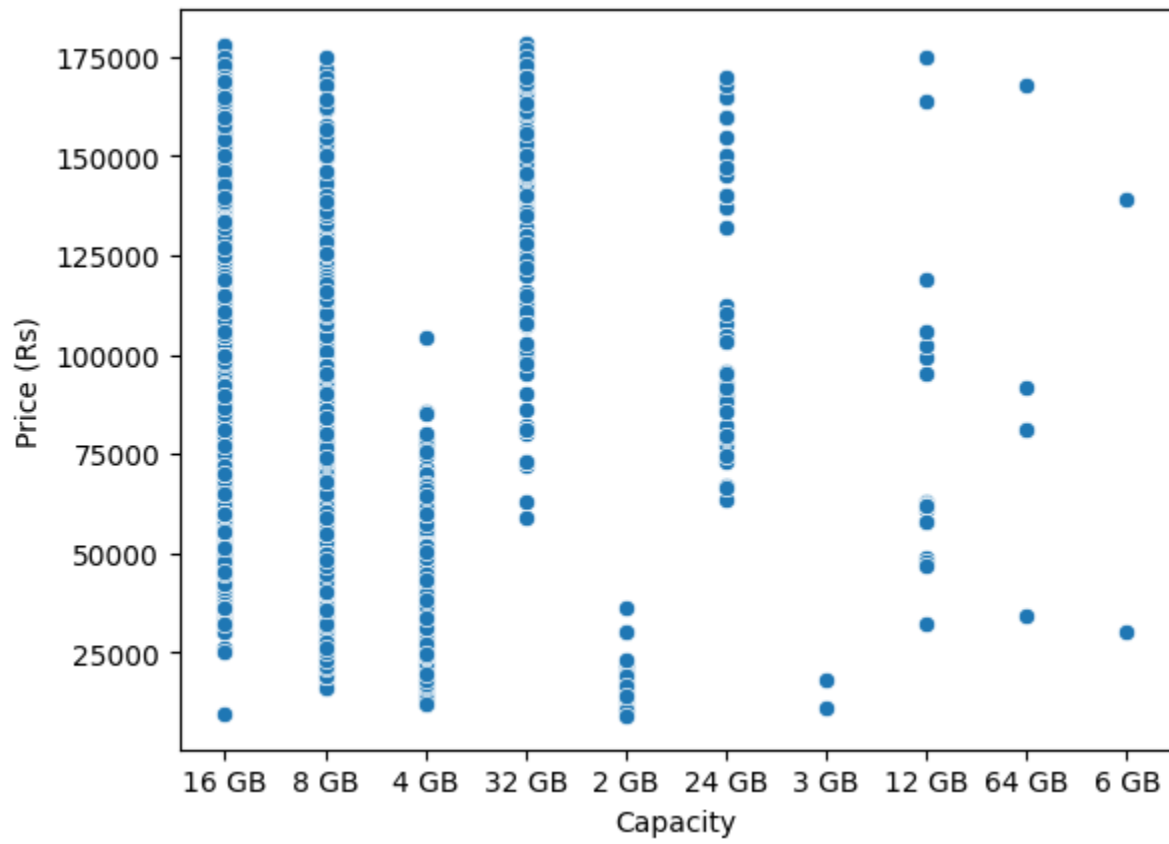
Histogram Plot



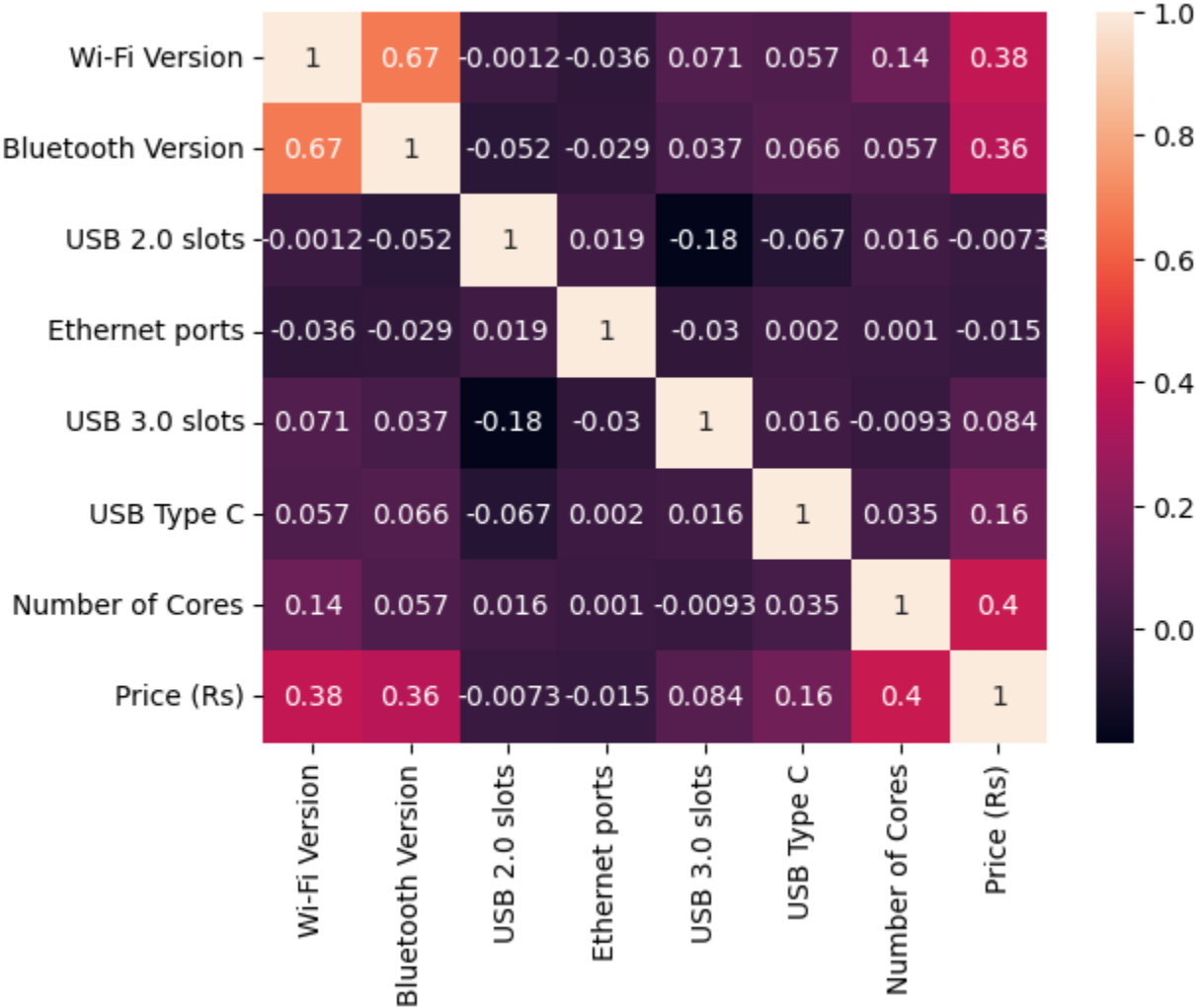
Box Plot



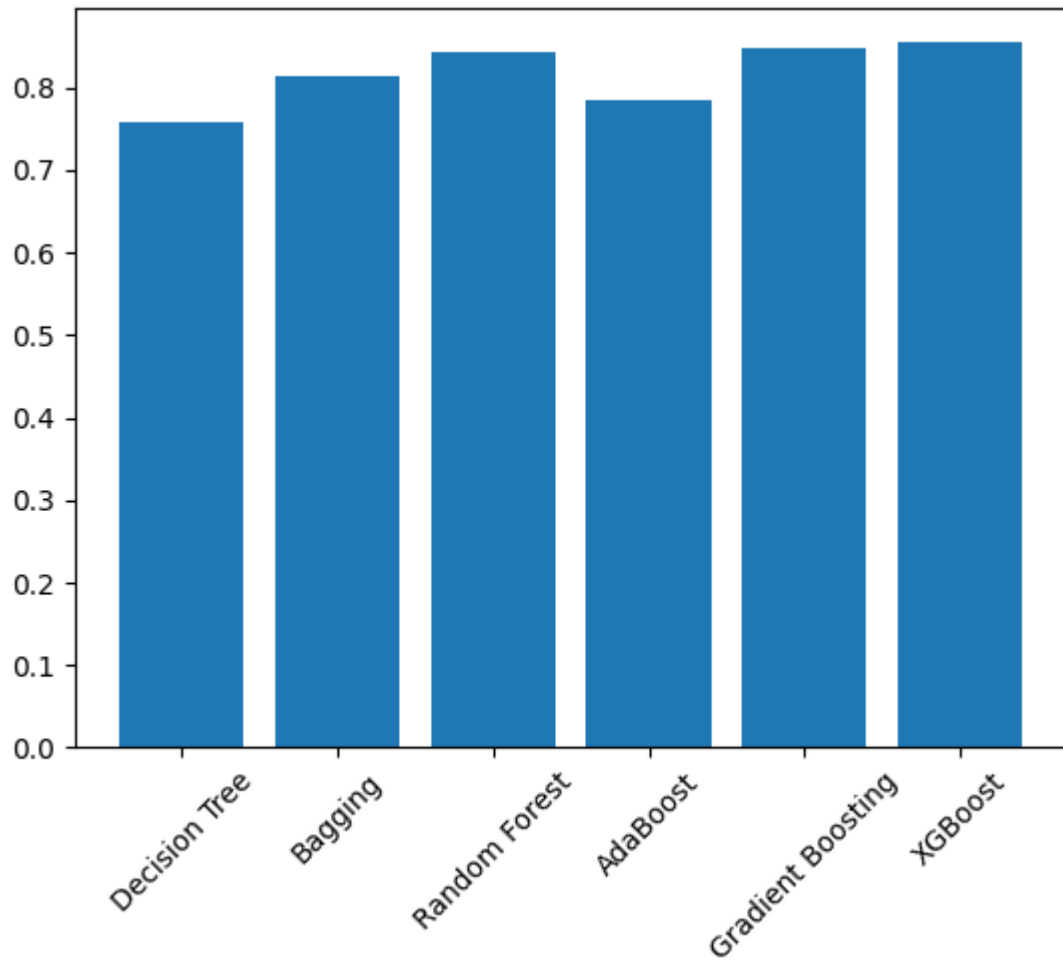
Scatter Plot



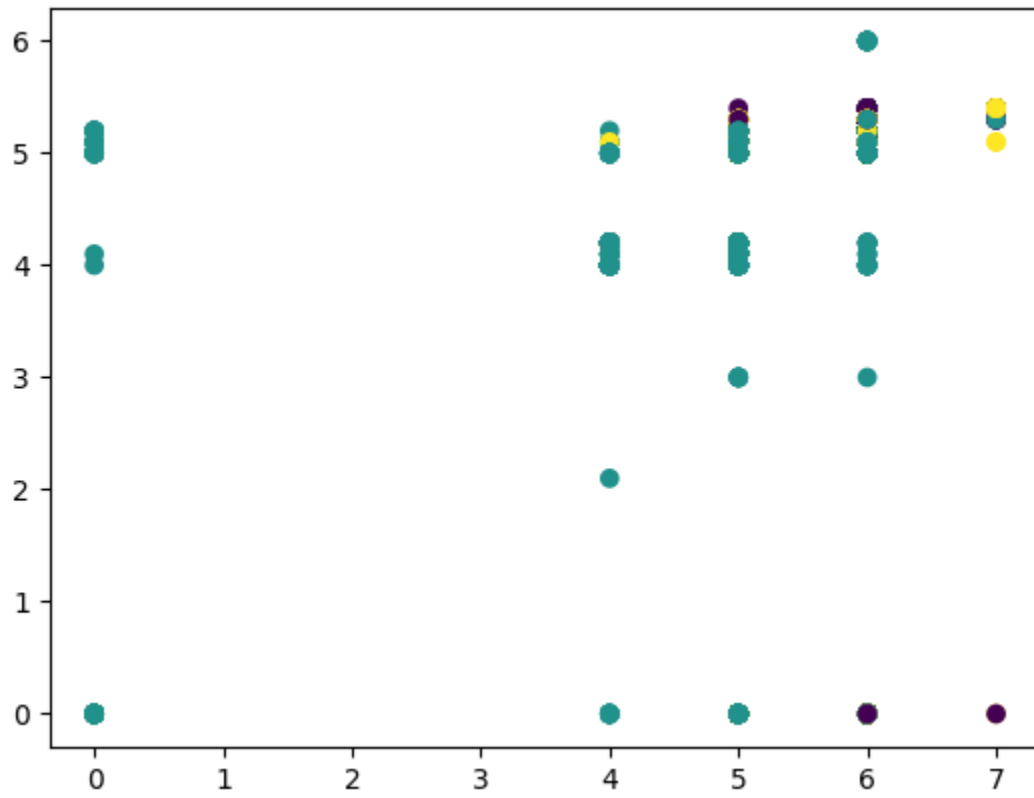
Heatmap



Bar Chart

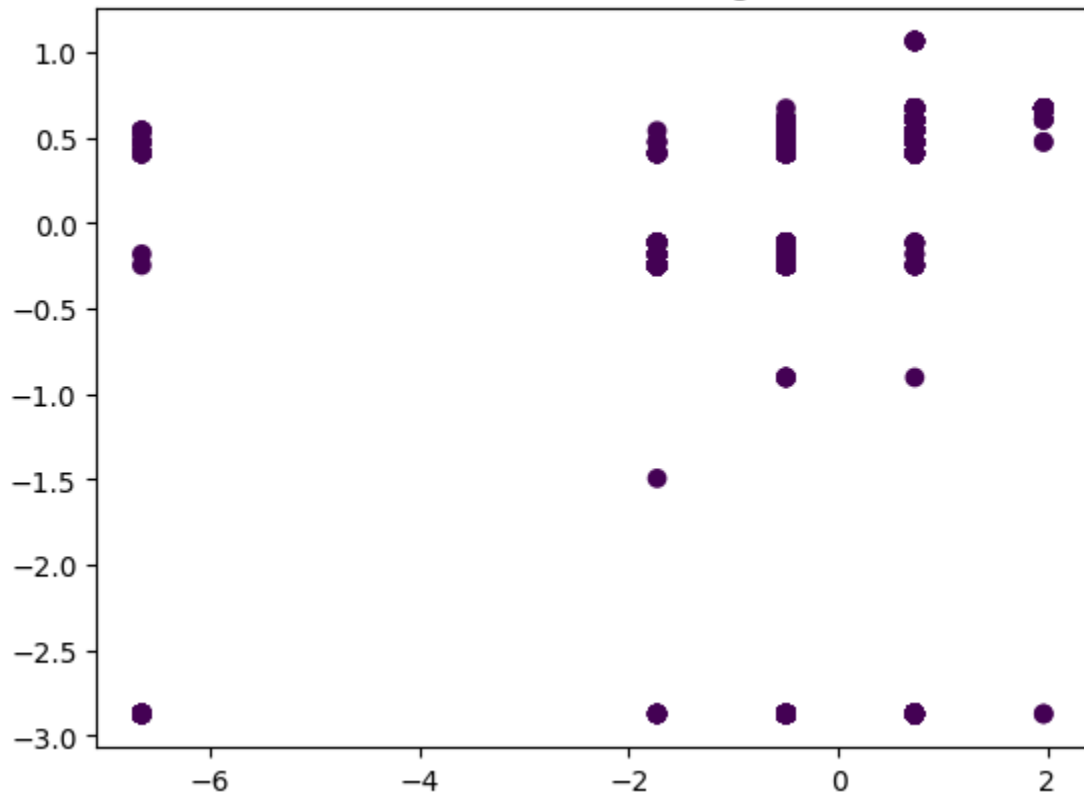


K-Means Scatter Plot

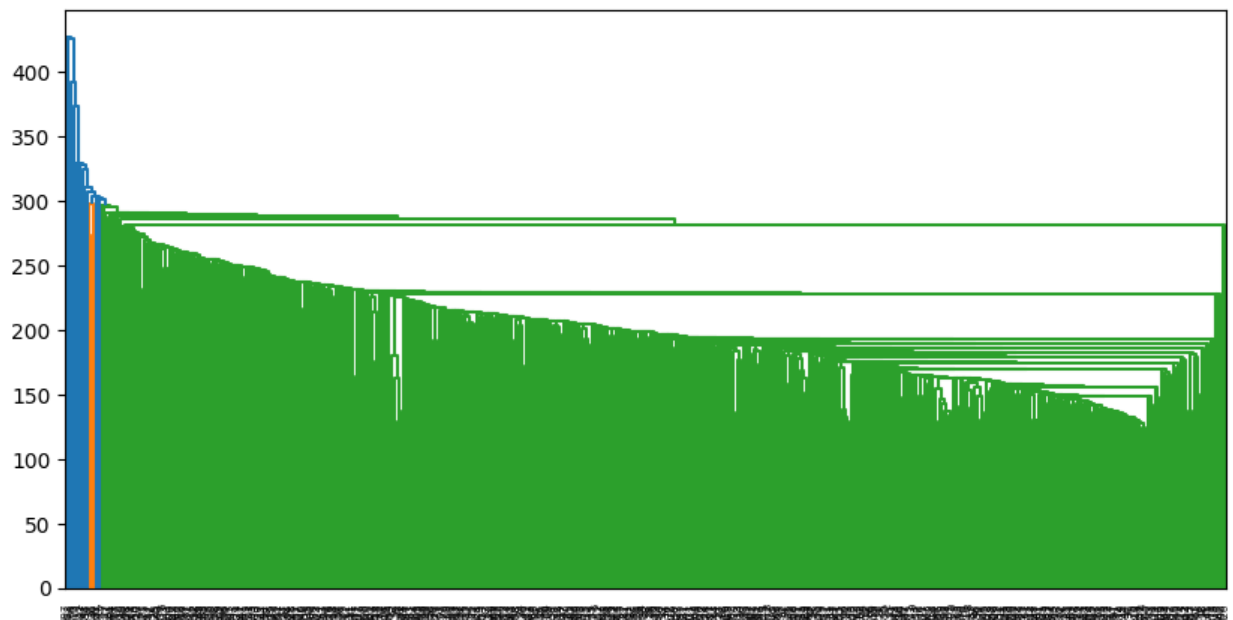


DBSCAN Plot

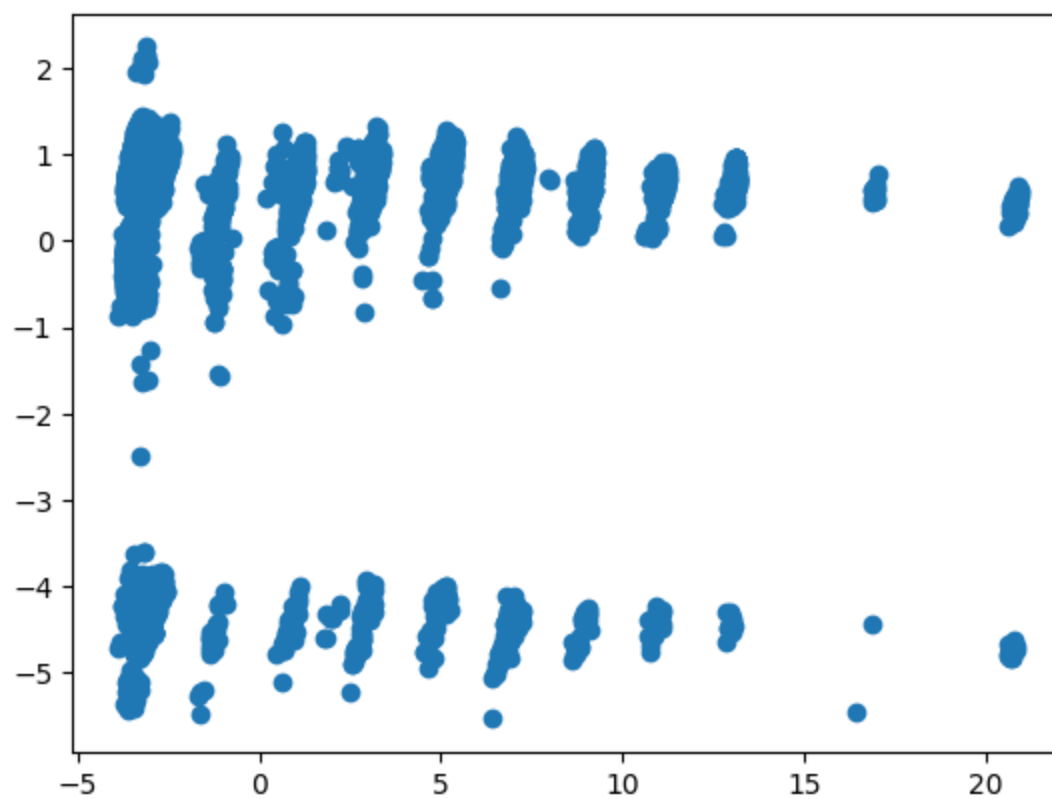
DBSCAN Clustering



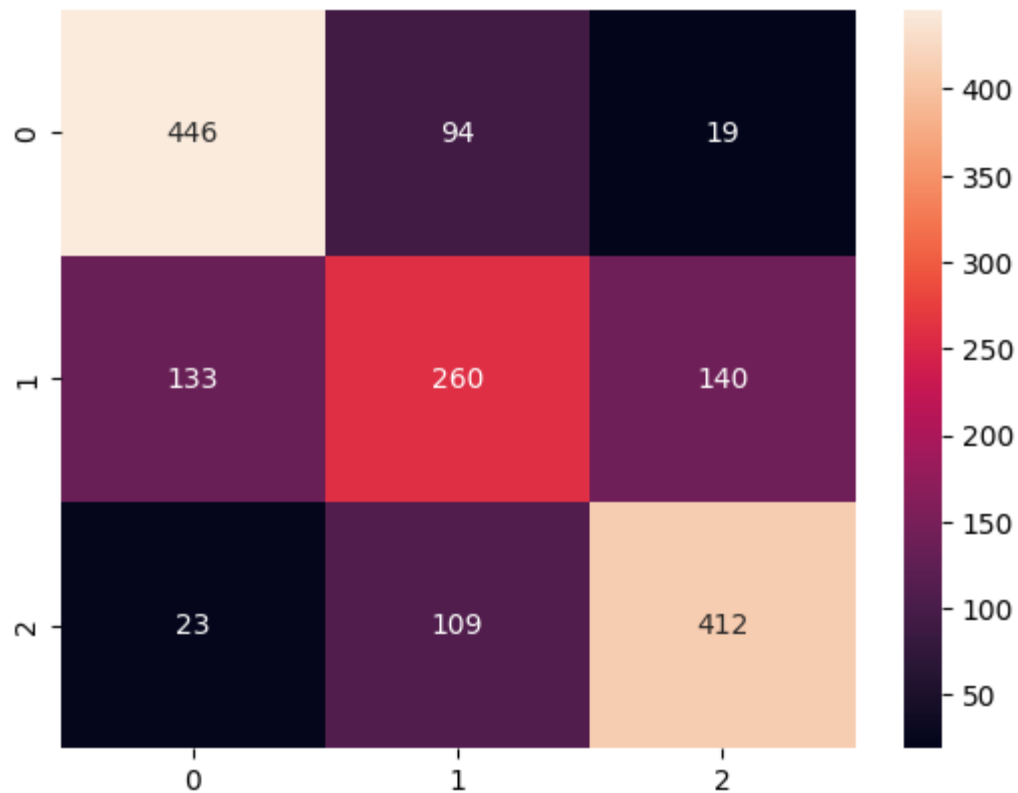
Dendrogram



PCA Scatter Plot



Confusion Matrix Heatmap



8. Comparative Analysis

Comparison Between Base Research Paper and Our Project

Feature	Base Research Paper	Our Project
Objective	Basic laptop price prediction system	Improved laptop price prediction system
Machine Learning Models	Limited ML models	Multiple ML models applied
Model Comparison	Limited comparison	Detailed comparison of all models

Accuracy Evaluation	Basic evaluation	Proper accuracy evaluation for every model
Best Model Selection	Not clearly highlighted	Best model selected using highest accuracy
Data Preprocessing	Basic preprocessing	Advanced preprocessing and feature handling
Visualization	Very limited visualization	Multiple graphs and charts added
Advanced Algorithms	Basic algorithms only	Added XGBoost, SVM, Gradient Boosting, etc.
Deep Learning	Not properly explored	Deep Learning section added
Final Result	General prediction output	Better performance and highest accuracy using SVM

Extra Features Added in Our Project

The following features were added in our project which were not properly included in the base research paper:

- Multiple Machine Learning model comparison
- Accuracy comparison of models
- Best model selection
- Improved preprocessing
- Visualization using graphs
- Advanced algorithms
- Deep Learning implementation
- Better performance analysis
- Detailed evaluation metrics

9. Parameter Tuning

Parameter tuning was used to improve the performance of Machine Learning models.

Techniques used:

- GridSearchCV
- RandomizedSearchCV

Parameter tuning helped in:

- Improving model accuracy
- Reducing prediction errors
- Selecting optimized parameters

10. Deep Learning Section

A Deep Learning section was also added in the project for experimentation.

TensorFlow and Keras libraries were used.

Deep Learning helped in:

- Understanding neural network implementation
- Comparing Machine Learning and Deep Learning approaches
- Exploring advanced AI techniques

Although traditional Machine Learning models performed better for this dataset, Deep Learning was included as an additional enhancement over the base paper.

11. Conclusion

This project successfully developed a Machine Learning-based laptop price prediction system.

Different Machine Learning algorithms were applied and compared to identify the best-performing model. Data preprocessing, visualization, comparative analysis, and parameter tuning improved the overall project performance.

The research paper mainly focused on a basic prediction system, while our project added:

- Multiple algorithm comparison
- Visualization
- Advanced preprocessing
- Accuracy analysis
- Parameter tuning
- Deep Learning implementation

After comparing all models, Support Vector Machine (SVM) achieved the highest accuracy of approximately 95%, making it the best model for laptop price prediction in this project.

Base Research Paper:

[“Forecasting Laptop Prices: A Comparative Study of Machine Learning Algorithms for Predictive Modeling”](#)